Describing Welfare

Session 2

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Constructing a consumption aggregate, and then describing the distribution of it, before modelling anything.

Reading

- Deaton, chs. 3 and 4
- Deaton & Zaidi, *Guidelines for Constructing Consumption Aggregates* — the operational reference, and the one you will actually reach for
- Deaton & Zaidi §§2–4 if you read nothing else

Why consumption, not income

- Income in an agricultural economy is *seasonal*, *lumpy*, and badly measured
  - "how much did you earn last year?" from a farmer with three crops
- Consumption is smoother, and closer to the theoretical object (permanent income, lifetime resources)
- Consumption is also easier to ask about: recall over short windows, item by item
- The cost: constructing it is real work. That work is this session.

The four components

A consumption aggregate is a sum of four things, in decreasing order of how well they are measured:

1. Food — purchased, own-produced, and received as gifts
2. Non-food, non-durable — fuel, transport, clothing, school fees, health
3. Durables — *use value*, not purchase price
4. Housing — rent, actual or imputed

Each of the last three involves an imputation, and each imputation is a place where two careful analysts will differ.

Own production and the valuation problem

Much of the food in an LSMS household was never bought.

- Value at what price?
  - farmgate price received (what the household gave up)
  - local market price (what it would have cost)
  - these differ by the marketing margin, which can be 30–50%
- Deaton & Zaidi: value at the price the household *faces*, consistently across households
- In practice: build a median unit value by item, by cluster, by round, and fall back up the hierarchy when cells are thin

Durables and housing

Durables: what enters consumption is the *flow of services*, not the purchase. For a stock S with service life L and interest rate r,

$$\text{use value} \;\approx\; S\left(r + \frac{1}{L}\right).$$

Housing: for owner-occupiers there is no rent, so impute it.

- hedonic regression of log rent on housing characteristics, among renters
- predict for owners
- fragile where the rental market is thin, which is most rural areas

Deflation

Two deflations, and they are different problems:

- Temporal: prices rise between rounds. Use the CPI, or better, a survey-based Paasche index built from the survey's own unit values.
- Spatial: prices differ across regions *within* a round, often by more than they differ across a decade. A national poverty line applied to undeflated nominal consumption will find poverty wherever prices are high.

Deaton & Zaidi recommend a household-specific Paasche index

$$P_i = \left[\sum_j w_{ij} \left(\frac{p_{ij}}{p_j^0} \right)^{-1} \right]^{-1},$$

with w_{ij} household i's budget share on item j.

Adult equivalence

A household of six is not six times as well off as a household of one with the same total consumption. Two adjustments, usually conflated:

- Household composition: children cost less than adults — $\Rightarrow$ adult-equivalent scale $A_i = \sum_k a_{\text{age,sex}}$
- Economies of scale: shared goods, so cost rises less than proportionately — $\Rightarrow n_i^\theta$, with $\theta \in (0, 1)$

Combined: $c_i = \dfrac{C_i}{A_i^\theta}$.

θ is not estimated in this session; it is *assumed*. Report the poverty rate for several values of it.

Poverty measures

The Foster–Greer–Thorbecke class: for poverty line z,

$$P_\alpha = \frac{1}{N} \sum_{i:c_i<z} \left(\frac{z - c_i}{z} \right)^\alpha .$$

- $\alpha = 0$: headcount ratio. Ignores *how* poor the poor are.
- $\alpha = 1$: poverty gap. The transfer needed, per capita, as a share of z.
- $\alpha = 2$: squared gap. Weights the poorest most; the only member of the family that satisfies the transfer axiom.

Weighted, always: $P_\alpha = \sum_i w_i \left(\frac{z-c_i}{z}\right)^\alpha / \sum_i w_i$.

Inequality

- Lorenz curve: cumulative share of consumption against cumulative share of population. Everything else is a summary of it.
- Gini: twice the area between the Lorenz curve and the 45° line. Bounded, familiar, and insensitive where it matters most.
- Generalized entropy $GE(\alpha)$: decomposable into within- and between-group parts. $GE(0)$ is mean log deviation, $GE(1)$ is Theil.

Prefer the whole Lorenz curve to any scalar. Prefer a decomposable scalar to the Gini when you want to say *where* inequality lives.

One index, or several?

Everything in this session divides nominal expenditure by a scalar: a Paasche index over space, a CPI over time, an equivalence scale over composition. Three scalars, one household, one number out.

That procedure is exactly right if the demand system has rank one.

- rank one means budget shares do not vary with total expenditure
- but this session opened by showing that they do — that is Engel's law
- so on what grounds is a single deflator adequate?

Hold the question. Session 3 estimates the rank for Uganda and answers it, and the answer is not reassuring.

Exercises

1. The poverty line used above is fake. Build a real one: take the food bundle consumed by households in the second and third deciles, price it at national median unit values, and scale it to 2,900 kcal per adult equivalent. Recompute P_0, P_1, P_2.
2. Deflate spatially. Construct a regional Paasche index from the survey's own unit values, apply it, and report how much of the north–south poverty gradient survives.
3. Vary θ in $c_i = C_i/A_i^{\theta}$ over $[0.5, 1.0]$ and plot the headcount ratio against it. Over what range does the *ranking* of regions change?